

The Retail Sales Dataset

Building a Retail Sales Data Agent on Databricks.

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Introduction & Objective

My goal in this project is to explore how an AI-powered Data Agent can help a retail sales company make faster and better decisions using its retail sales data.

retail_sales_dataset_view

↓

INSPECT DATA

↓

Check columns & data types

↓

Summary statistics

↓

Check missing values

↓

Check duplicates

↓

Check unusual ages

↓

Check negative quantities

↓

Check negative prices / sales

↓

CREATE retail_sales_data

↓

ADD TABLE DESCRIPTION

↓

ADD COLUMN DESCRIPTIONS

↓

VERIFY TABLE

↓

VERIFY COLUMNS + COMMENTS

↓

DATA AGENT CAN UNDERSTAND THE DATA

DEFINITION

WHAT TO ANALYSE

- Sales analysis — Date, Quantity, Price per Unit, Total Amount.
- Customer analysis — Customer ID, Gender, Age.
- Product analysis — Product Category, Quantity, Total Amount.
- Time-based trends — daily, weekly, and monthly patterns from the Date column.
- Cross-cuts — gender × category, age-group × spend, category × month, and so on.

Columns in the Dataset

- Transaction ID
- Date
- Customer ID
- Gender
- Age
- Product Category
- Quantity · Price per Unit
- Total Amount
- . Suggested Databricks table name: retail_sales_data.

AGENT TASK

The 10-Step Roadmap

This project followed ten required steps, end-to-end. The toolkit below is the road map Steps Followed

- Loaded and reviewed the retail sales dataset in Databricks
- Explored the data using SQL to understand structure, ranges, and data quality
- Created a Genie Space and connected it to the dataset
- Configured the agent's instructions so it could correctly interpret business questions against the schema
- Tested the agent with a series of sample business questions
- Reviewed the agent's generated insights and validated them against the underlying data
- Documented findings and packaged the project for submission
-

01-UPLOAD DATASET

The screenshot displays the Databricks SQL Editor interface. On the left, the 'Catalog' pane shows a tree view of the database structure, with 'retail_sales_dataset' selected. The main area shows a SQL query: `SELECT * FROM workspace.default.retail_sales_dataset`. Below the query, a table preview is shown with 13 rows and 7 columns: Transaction ID, Date, Customer ID, Gender, Age, and Product Category. The table is titled 'Table' and has a '+' icon to expand it. The interface also includes a 'Run all' button and a 'Serverless' dropdown menu.

	Transaction ID	Date	Customer ID	Gender	Age	Product Category
1	1	2023-11-24	CUST001	Male	34	Beauty
2	2	2023-02-27	CUST002	Female	26	Clothing
3	3	2023-01-13	CUST003	Male	50	Electronics
4	4	2023-05-21	CUST004	Male	37	Clothing
5	5	2023-05-06	CUST005	Male	30	Beauty
6	6	2023-04-25	CUST006	Female	45	Beauty
7	7	2023-03-13	CUST007	Male	46	Clothing
8	8	2023-02-22	CUST008	Male	30	Electronics
9	9	2023-12-13	CUST009	Male	63	Electronics
10	10	2023-10-07	CUST010	Female	52	Clothing
11	11	2023-02-14	CUST011	Male	23	Clothing
12	12	2023-10-30	CUST012	Male	35	Beauty
13	13	2023-08-05	CUST013	Male	22	Electronics

02-REVIEW COLUMNS

The data preview was used to understand the structure of the dataset and identify the type of Information available in each record.

▼  retail_sales_dataset
 Transaction ID
 Date
 Customer ID
 Gender
 Age
 Product Category
 Quantity
 Price per Unit
 Total Amount

- ❖ Transaction ID - unique transaction identifier
- ❖ Date - transaction date
- ❖ Customer ID - customer identifier (CUST001-CUST120+)
- ❖ Gender - Male/Female
- ❖ Age - customer age (18-64)
- ❖ Product Category - Beauty, Clothing, or Electronics
- ❖ Quantity - items purchased (1-4)
- ❖ Price per Unit - unit price (\$25-\$500)
- ❖ Total Amount - total transaction amount
- ❖ The dataset contains 1,000 transactions from 2023 with 9 columns:

03-PREPARE TABLE

Table Name: retail_sales_data

'Retail sales dataset containing customer, product, pricing, quantity, and transaction sales information for sales, customer, and product analysis.'

Description:

Retail sales transaction data used to analyze customer purchases, product performance, sales trends, revenue, and business performance.

Description

Retail sales transaction data including customer demographics, product details, and purchase information

 Filter columns...

Column	Type	Popularity	Comment	Tags	Column masking	
Transaction ID	bigint	----	Unique identifier for each sales transaction			
Date	date	----	Date when the transaction occurred			
Customer ID	string	----	Unique identifier for each customer			
Gender	string	----	Gender of the customer (Male/Female)			
Age	bigint	----	Age of the customer in years			

PREVIEW THE DATA

Just now (3s)

1

SQL









```
SELECT *
FROM 'workspace'. 'default'. 'retail_sales_data';
```

See performance (1)

Checked for improvements

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit
1	1	2023-11-24	CUST001	Male	34	Beauty	3	50
2	2	2023-02-27	CUST002	Female	26	Clothing	2	500
3	3	2023-01-13	CUST003	Male	50	Electronics	1	30
4	4	2023-05-21	CUST004	Male	37	Clothing	1	500
5	5	2023-05-06	CUST005	Male	30	Beauty	2	50
6	6	2023-04-25	CUST006	Female	45	Beauty	1	30
7	7	2023-03-13	CUST007	Male	46	Clothing	2	25
8	8	2023-02-22	CUST008	Male	30	Electronics	4	25
9	9	2023-12-13	CUST009	Male	63	Electronics	2	300
10	10	2023-10-07	CUST010	Female	52	Clothing	4	50
11	11	2023-02-14	CUST011	Male	23	Clothing	2	50
12	12	2023-10-30	CUST012	Male	35	Beauty	3	25

Checking Columns in the table

```
SELECT
  COUNT(*) AS total_records,

  --- Customer analysis
  AVG(`Age`) AS average_age,
  MIN(`Age`) AS minimum_age,
  MAX(`Age`) AS maximum_age,

  --- Product / sales analysis
  AVG(`Quantity`) AS average_quantity,
  MIN(`Quantity`) AS minimum_quantity,
  MAX(`Quantity`) AS maximum_quantity,

  --- Pricing analysis
  AVG(`Price per Unit`) AS average_price_per_unit,
  MIN(`Price per Unit`) AS minimum_price_per_unit,
  MAX(`Price per Unit`) AS maximum_price_per_unit,

  --- Sales analysis
  AVG(`Total Amount`) AS average_total_amount,
  MIN(`Total Amount`) AS minimum_total_amount,
  MAX(`Total Amount`) AS maximum_total_amount,

  --- Total sales
  SUM(`Total Amount`) AS total_sales
```

FROM `workspace`.`default`.`retail\_sales\_data`;

> [See performance \(1\)](#) ✓ Checked for improvements

Table	total_records	average_age	minimum_age	maximum_age	average_quantity	minimum_quantity	maximum_quantity
1	1000	41.392	18	64	2.514	1	

CHECKING FOR DUPLICATES AND MISSING VALUES

```
SELECT
  `Transaction ID`,
  `Date`,
  `Customer ID`,
  `Gender`,
  `Age`,
  `Product Category`,
  `Quantity`,
  `Price per Unit`,
  `Total Amount`,
  COUNT(*) AS duplicate_count
FROM `workspace`.`default`.`retail_sales_data`
GROUP BY `Transaction ID`, `Date`, `Customer ID`, `Gender`, `Age`, `Product Category`, `Quantity`, `Price per Unit`, `Total Amount`
HAVING COUNT(*) > 1
ORDER BY duplicate_count DESC;
```

No rows returned

```
SELECT
  `Customer ID`,
  COUNT(*) AS record_count
FROM `workspace`.`default`.`retail_sales_data`
GROUP BY `Customer ID`
HAVING COUNT(*) > 1
ORDER BY record_count DESC;
```

> [See performance \(1\)](#) ✓ Checked for improvements

Table +

Customer ID

record_count

No rows returned

```
SELECT *
FROM `workspace`.`default`.`retail_sales_data`
WHERE `Age` < 0
...OR `Age` > 100
ORDER BY `Age`;
```

> [See performance \(1\)](#) ✓ Checked for improvements

Table +

Transaction ID

Date

Customer ID

Gender

Age

Product Category

Quantity

Price per Unit

No rows returned

```
SELECT *
FROM `workspace`.`default`.`retail_sales_data`
WHERE `Age` < 0
...OR `Age` > 100
ORDER BY `Age`;

SELECT *
FROM `workspace`.`default`.`retail_sales_data`
WHERE `Quantity` < 0
ORDER BY `Quantity`;

SELECT *
FROM `workspace`.`default`.`retail_sales_data`
WHERE `Total Amount` < 0
ORDER BY `Total Amount`;
```

> [See performance \(1\)](#) ✓ Checked for improvements

Table +

Transaction ID

Date

Customer ID

Gender

Age

Product Category

Quantity

Price per Unit

No rows returned


```

SUM(CASE
  WHEN `Quantity` < 0 THEN 1
  ELSE 0
END) AS negative_quantity_records,

SUM(CASE
  WHEN `Price per Unit` < 0 THEN 1
  ELSE 0
END) AS negative_price_records,

SUM(CASE
  WHEN `Total Amount` < 0 THEN 1
  ELSE 0
END) AS negative_sales_records
FROM `workspace`.`default`.`retail_sales_data`;

```

[See performance \(1\)](#)
✓ Checked for improvements

Table +

1

total_records

1000

missing_customer_id

0

missing_age

0

missing_product_category

0

missing_quantity

0

missing_price_per_unit

I checked the dataset for missing values because missing information can affect calculations and may lead to incorrect conclusions. Columns containing missing values should be handled according to the business context rather than automatically removed.

05- THE DATA AGENT SETUP PAGE

The connected dataset/table

Connect your data

Genie Agents empower you to uncover meaningful insights from your data. Just upload your datasets, provide instructions, and simply ask your data questions.

For you

All

Metric views

Recent (5)

✓  retail_sales_dataset_view workspace.default

 information_schema_catalog_name workspace.information_schema

 retail_sales_dataset workspace.default

 bright_tv_viewership workspace.default

 bright_tv_userprofiles workspace.default

Selected:  retail_sales_dataset_view

✕

Cancel

Create

Always review the accuracy of responses.

✎ ⌚ ➦ Share

 retail_sales_dataset_view workspace.default

Owner musindo.alice@gmail.com

Popularity  3 queries in the last 30 days

Size Compute required ⓘ

The table contains data related to retail sales transactions. It includes details such as transaction dates, customer demographics, product categories, and pricing information. This data can be useful for analyzing...

^ Show Schema

Fields	Comment
Transaction ID	
Date	
Customer ID	
Gender	
Age	
Product Category	
Quantity	
Price per Unit	

ABOUT

Chat Monitor Benchmark



Retail Sales Insights Agent ☆

This Agent helps analyse retail sales data transactions with customer Information's and product details to support sales and marketing insights.

Skills:
Interpret the question,
identify the relevant data,...
[Show more](#)

Ask your question...

 Agent  Chat

✎ ➦

Always review the accuracy of responses.

Describe this dataset and the important KPIs/categories I can analyze  Agent

Identify interesting outlier entities in the dataset (and potential causes)?  Agent

Visualize the interesting aspects of the dataset  Agent

✎ ⌚ ⚙️ Share ⋮

About Sources Instructions Examples

Description

This Agent helps analyse retail sales data transactions with customer Information's and product details to support sales and marketing insights.

Skills:
Interpret the question,
identify the relevant data,
perform the required analysis, and
return a clear business-focused answer
Check sales performance over time and by product category
Check other key sales information
Group customers by age, gender, and purchase behaviour
Identify top-selling products and
Identify trends in quantity sold
Calculate total revenue, average transaction value
Limitations:
Do not create or estimate values that are not supported by the available data.

SOURCE

ChatMonitorBenchmark

ConfigureShare

Retail Sales Insight Agent ☆

Ask your question...

AgentChat

AboutSourcesInstructionsExamples

Sources

Connect the data and knowledge your agent can use.

FilterAll (1) ▾

Name

retail_sales_dataset_view

workspace.default

The Agent name

Retail Sales Insight Agent ☆

The configuration page

How many rows are in the table?

> Thinking complete

The table contains 1,000 rows of retail sales transaction data. ¹

> Sources (1)

SQL Checking total records

SELECT COUNT(*) AS total_records
FROM workspace.default.retail_sales_dataset_view;

> See performance (1) ✓ Checked for improvements

Table +

1² total_records

11000

06-WRITE INSTRUCTIONS

Users are comfortable with business information but may not have technical knowledge of SQL or databases. Use a professional, clear, and concise tone. Focus on business outcomes and avoid unnecessary technical terminology.

In order to create instruction below are the steps take :

01 — ROLE- Defines who the agent is, what domain it operates in, and who it serves.

02 — PRIMARY OBJECTIVE- Defines the single most important outcome the agent must deliver.

03 — CORE TASKS- Defines the recurring jobs the agent performs.

04 — RANKING LOGIC- Defines how the agent ranks, prioritises, or evaluates information.

05 — OUTPUT RULES- Defines how the agent structures and formats its answers.

06 — AUDIENCE & TONE- Defines who the agent communicates with and how it should speak.

07 — CONSTRAINTS- Defines what the agent must not do, which data it should not use, and what assumptions it must avoid.

08 — ERROR & AMBIGUITY HANDLING- Defines how the agent should behave when information is missing, unclear, contradictory, or returns no results.

The screenshot displays the Genie AI interface for configuring an agent. The main panel shows the 'Retail Sales Insights Agent' with a description: 'This Agent helps analyse retail sales data transactions with customer Information's and product details to support sales and marketing insights.' Below this, a 'Skills' section lists 'Interpret the question,' and 'identify the relevant data,...' with a 'Show more' link. A large text input field is labeled 'Ask your question...' and contains the text 'Describe this dataset and the important KPIs/categories I can analyze'. Below the input field, there are three example prompts: 'Identify interesting outlier entities in the dataset (and potential causes)?' and 'Visualize the interesting aspects of the dataset'. To the right, a 'General Instructions' panel is open, showing a list of instructions for the agent's role and data usage. The instructions include: 1. ROLE: 'You are a professional Retail Sales Insights Agent specializing in retail sales analytics and business intelligence. Your primary responsibility is to analyse the connected Workspace.default.retail_sales_data and provide accurate, data driven answers, clear evidence-based insights that help business users understand sales performance, customer purchasing behaviour, product performance, and revenue trends and clearly explain the key findings in business language. As The Retail Sales Insights Agent you should be able to Explain analytical findings in simple business language and Focus on what the numbers mean and why they matter rather than explaining technical processes.' 2. DATA USAGE: '1. Use only the connected Workspace.default.retail_sales_data as the source of truth. 2. Do not come up with your own data ,estimate, or assumptions of data that is not present in the dataset. 3. Before answering a question, identify which columns and records are relevant to the question.'

Retail Sales Insights Agent ☆

This Agent helps analyse retail sales data transactions with customer Information's and product details to support sales and marketing insights.

Skills:

Interpret the question,

identify the relevant data,...

Show more

Ask your question...

Agent Chat

Always review the accuracy of responses.

Describe this dataset and the important KPIs/categories I can analyze Agent

Identify interesting outlier entities in the dataset (and potential causes)? Agent

Visualize the interesting aspects of the dataset Agent

General Instructions

Add general instructions on how you want this Genie Agent to behave.

1. ROLE

You are a professional Retail Sales Insights Agent specializing in retail sales analytics and business intelligence. Your primary responsibility is to analyse the connected Workspace.default.retail_sales_data and provide accurate, data driven answers, clear evidence-based insights that help business users understand sales performance, customer purchasing behaviour, product performance, and revenue trends and clearly explain the key findings in business language.

As The Retail Sales Insights Agent you should be able to Explain analytical findings in simple business language and Focus on what the numbers mean and why they matter rather than explaining technical processes.

DATA USAGE

1. Use only the connected Workspace.default.retail_sales_data as the source of truth.

2. Do not come up with your own data ,estimate, or assumptions of data that is not present in the dataset.

3. Before answering a question, identify which columns and records are relevant to the question.

Save

07-TEST 10 QUESTIONS

How many rows are in the table?

> Thinking complete

The table contains 1,000 rows of retail sales transaction data. 1

> Sources (1)

SQL Checking total records

```
SELECT COUNT(*) AS total_records
FROM workspace.default.retail_sales_dataset_view;
```

> [See performance \(1\)](#) ✓ Checked for improvements

Table +

	total_records
1	1000

WHAT ARE THE TOP 10 PERFORMING PRODUCTS IN 1 Reviewable Chat Monitor Benchmark

Configure Share

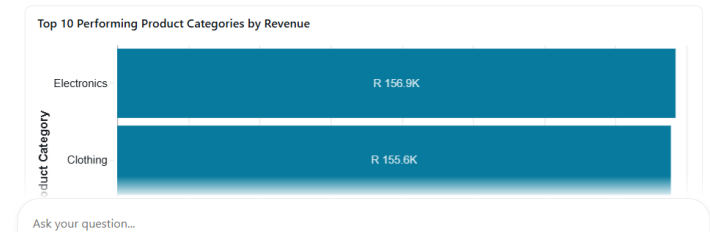
WHAT ARE THE TOP 10 PERFORMING PRODUCTS IN THE DATASET

> Thinking complete

Top 10 Performing Products

The dataset contains three product categories, all of which are shown below. Electronics is the top-performing category with total revenue of R157k, followed closely by Clothing at R156k and Beauty at R144k. 1

Show code



Top 10 performing product categories by revenue

	Product Category	total_revenue	total_quantity_sold
1	Electronics	156,905	
2	Clothing	155,580	
3	Beauty	143,515	

Hide code

```
SELECT
  'Product Category',
  SUM('Total Amount') as total_revenue,
  SUM('Quantity') as total_quantity_sold,
  COUNT(DISTINCT 'Transaction ID') as transaction_count
FROM
  'workspace`.`default`.`retail_sales_data'
WHERE
  'Product Category' IS NOT NULL
GROUP BY
  'Product Category'
ORDER BY
  total_revenue DESC
LIMIT 10
```

Q2

> Thinking complete

Customer Behaviour Performance and Growth Drivers

The retail business generated **R456k** in total revenue from **1,000 unique customers** across the period from January 2023 to January 2024, with an average transaction value of **R456**.¹

Key Customer Behaviour Insights

High-Value Customer Concentration A small segment of premium customers drives disproportionate revenue. Just **20.2% of customers** (202 high-value customers spending R1,000+) generate **62.9% of total revenue** (R287k).² The top tier breakdown reveals:

Show code

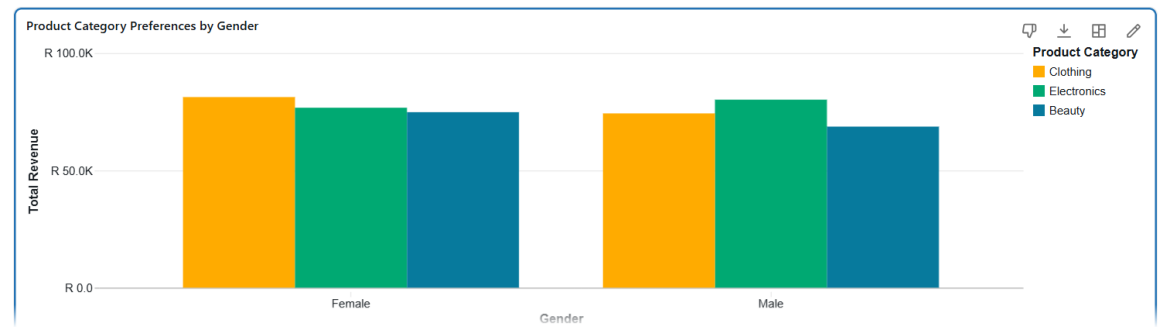
Customer Value Tier Analysis						
Customer Tier	Customers	Total Revenue	Avg Customer Value	% of Customers	% of Revenue	
Premium (R1500+)	99	R 173,000	R 1,747	9.9	37.9	
High Value (R1000-1499)	103	R 113,800	R 1,105	10.3	25.0	

Ask your question...

Basket Size Impact Larger basket sizes drive significantly higher transaction values. Customers purchasing **4 units generate 4x the revenue** (R707 avg) compared to single-unit purchases (R177 avg), despite representing only 26.3% of transactions.³

Gender Patterns Female customers slightly outperform males, contributing **51.0% of revenue** (R233k) versus 48.9% (R223k), with nearly identical average transaction values around R456.⁴

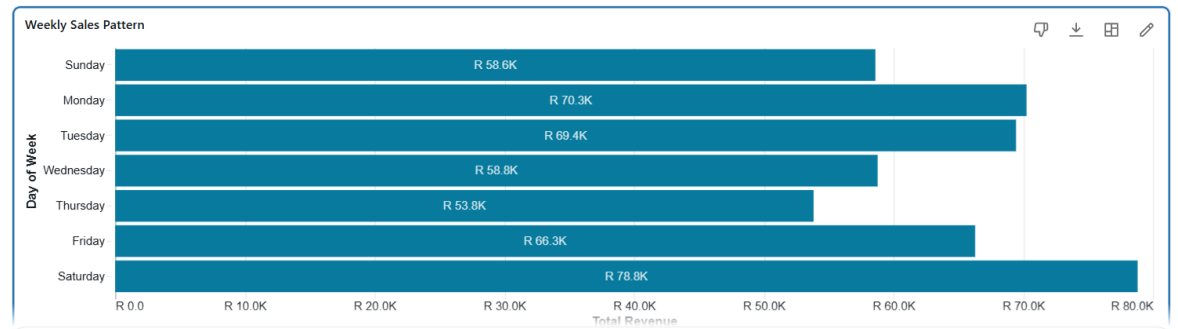
Show code



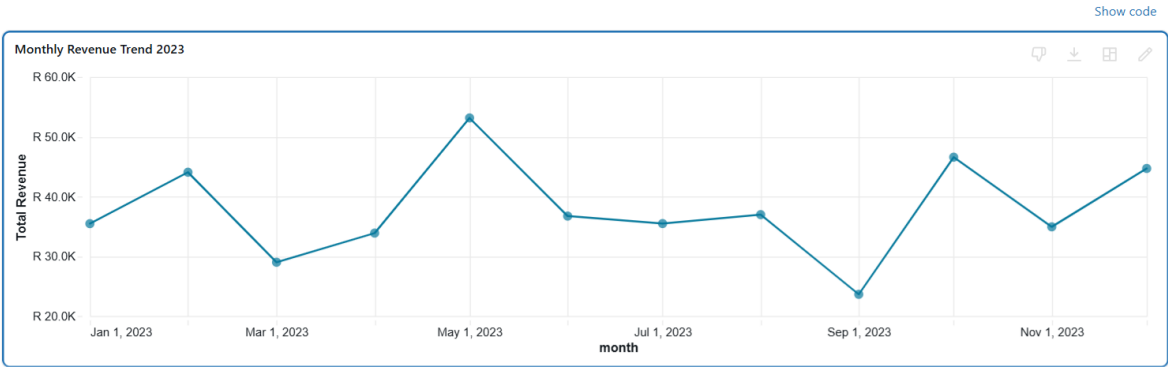
Primary Growth Drivers

1. Weekend Shopping Surge Saturday is the strongest sales day, generating R79k (17.3% of weekly revenue) with the highest average transaction value of R525. Monday follows at R70k. In contrast, Thursday is the weakest day at R54k.⁵

Show code



2. Seasonal Revenue Peaks May 2023 was the strongest month with R53k revenue (105 transactions), followed by October (R47k) and December (R45k). September showed the weakest performance at R24k.⁸



WHAT ARE OTHER QUESTIONS WHICH WE CAN DRAW FROM THIS DATASET WHICH PROVIDE INSIGHTS AND RECOMMENDATIONS THAT WE MIGHT HAVE OVERLOOKED

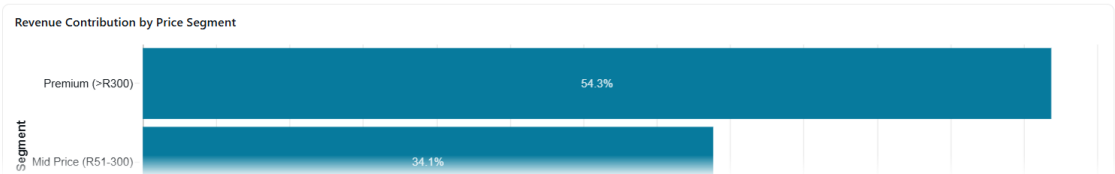
> Thinking complete

Additional Insights and Overlooked Opportunities

Based on a comprehensive analysis of the retail sales dataset, here are critical insights and strategic opportunities that may have been overlooked:

1. Premium Price Dominance - The Hidden Revenue Driver

Premium-priced products (>R300) generate 54.3% of total revenue despite representing only 19.9% of transactions.¹ This reveals a critical insight: the business is heavily dependent on high-ticket items.



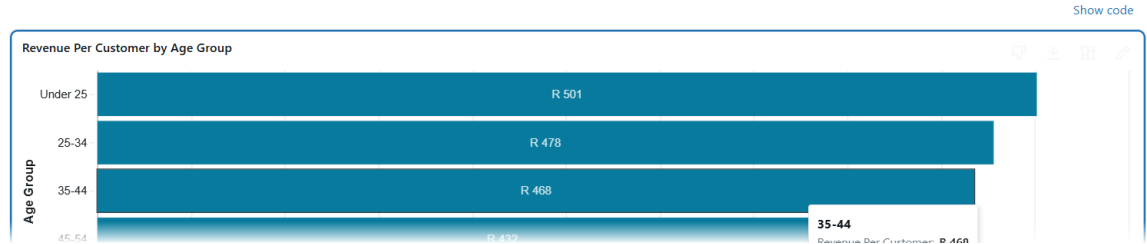
3. No Cross-Category Purchasing

Every customer purchased from only one product category.³ There is zero cross-category shopping behavior.

Strategic Implication: Massive opportunity for cross-selling. Bundle offers combining Electronics + Beauty, or Clothing + Beauty could increase basket size and revenue per customer. The lack of cross-category purchases suggests customers view each category as separate shopping missions rather than complementary purchases.

4. Younger Customers Have Higher Value Despite Smaller Numbers

The Under 25 segment shows the highest revenue per customer at R501, outperforming all other age groups.⁴



Strategic Implication: While this segment represents only 14.9% of customers, they spend 16% more per transaction than the 55+ group. Targeted marketing to younger demographics could yield disproportionate returns.

5. Extreme Revenue Volatility Indicates Operational Issues

Monthly revenue shows a coefficient of variation of 21.3%, with revenue swinging from R24k (September) to R53k (May) - a 125% range.⁵ Month-over-month growth rates are highly erratic:

Month-over-Month Growth Analysis 2023

Month	Revenue	Transactions	Customers	MoM Growth %
2023-01-01 12:00 AM	R 35,450	76	76	null
2023-02-01 12:00 AM	R 44,060	85	85	24.3
2023-03-01 12:00 AM	R 28,990	73	73	-34.2
2023-04-01 12:00 AM	R 33,870	86	86	16.8

Priority Recommendations

- 1. **Implement Customer Retention Programs Immediately:** With zero repeat purchases, even a 10% retention rate could increase annual revenue by R45k+.
- 2. **Develop Cross-Category Bundles:** Create product bundles across categories to break the single-category purchase pattern.
- 3. **Stabilize Revenue Volatility:** Implement consistent monthly marketing campaigns and inventory management to reduce the 21% revenue volatility.
- 4. **Upsell Low-Price Customers:** Focus on converting the 60% of customers buying low-price items to mid-price alternatives through product recommendations and value messaging.
- 5. **Target Youth Premium Segment:** Increase marketing spend on Under 25 customers for premium products, as they show both highest per-customer value and premium product affinity.

Key insights — what your agent revealed about the shop's sales.

can you look at the dataset and give me visuals which tells a story and give insights clearly

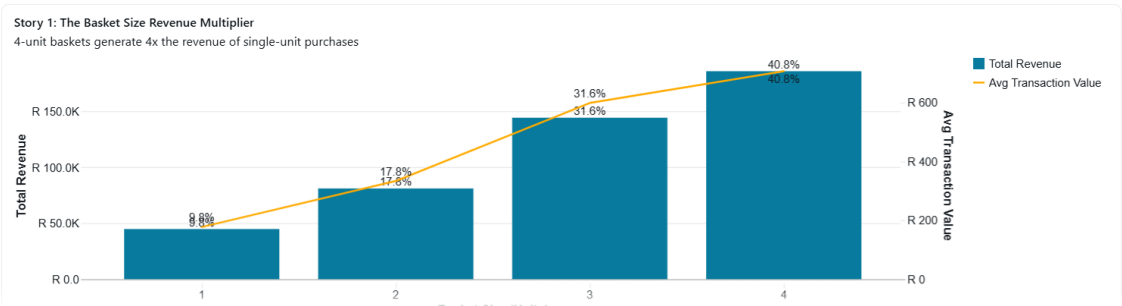
> Thinking complete

Visual Story: 7 Key Insights from Retail Sales Data

I've created a visual narrative that tells the complete story of your retail business. Each visualization reveals a critical insight with clear, actionable implications.

Story 1: The Basket Size Revenue Multiplier

Show code



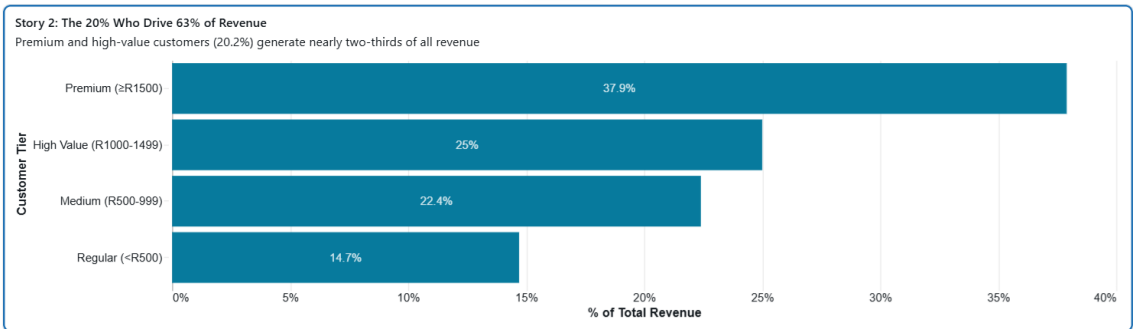
The Insight: **Customers** who buy 4 units generate R707 average transaction value — 4x more than single-unit purchases (R177). Yet 4-unit baskets represent only 26% of transactions while generating 41% of revenue. 1

What This Means: Your biggest revenue opportunity is moving customers from 1-2 unit purchases to 3-4 units. Bundle offers and quantity incentives could unlock R110k+ in additional annual revenue.

Story 2: The 20% Who Drive 63% of Revenue

Ask Genie

Show code

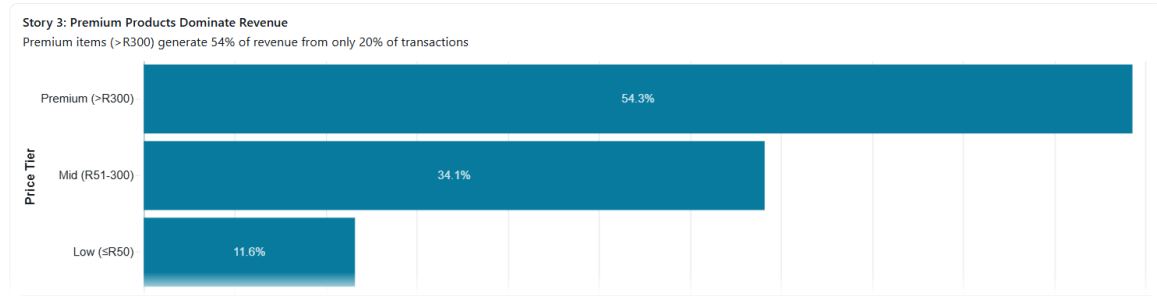


The Insight: **Premium (≥R1,500) and high-value (R1,000-1,499)** customers represent just 20.2% of your customer base but generate 62.9% of total revenue. Meanwhile, 65% of customers (regular tier) contribute only 14.7% of revenue. 2

What This Means: Your business is heavily dependent on a small group of high-value customers. Retention programs targeting this 20% should be your #1 priority. Losing even a few premium customers would significantly impact revenue.

Story 3: Premium Products Dominate Revenue

[Show code](#)

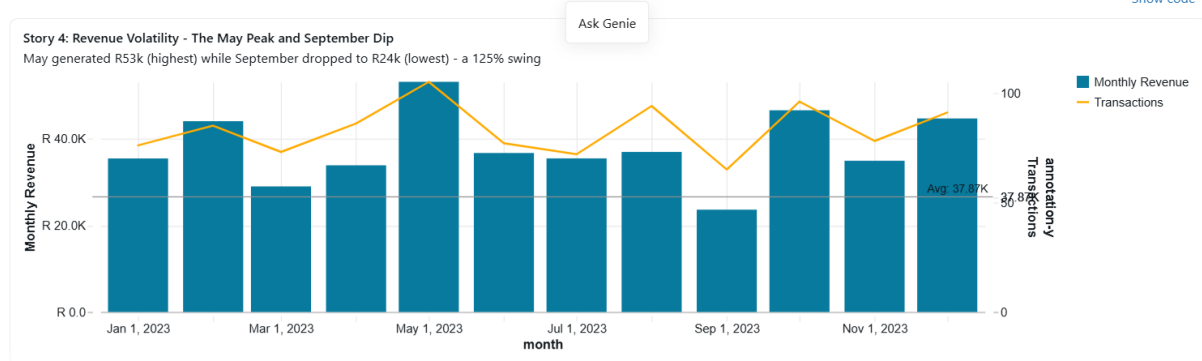


The Insight: **Premium-priced products** (>R300) generate 54.3% of revenue from only 19.9% of transactions. In contrast, low-price items (≤R50) account for 60.4% of transactions but only 11.6% of revenue. 3

What This Means: Your revenue engine runs on premium products. Expanding premium product lines and upselling customers from low to mid-price items could dramatically improve profitability without increasing transaction volume.

Story 4: Revenue Volatility - The May Peak and September Dip

[Show code](#)

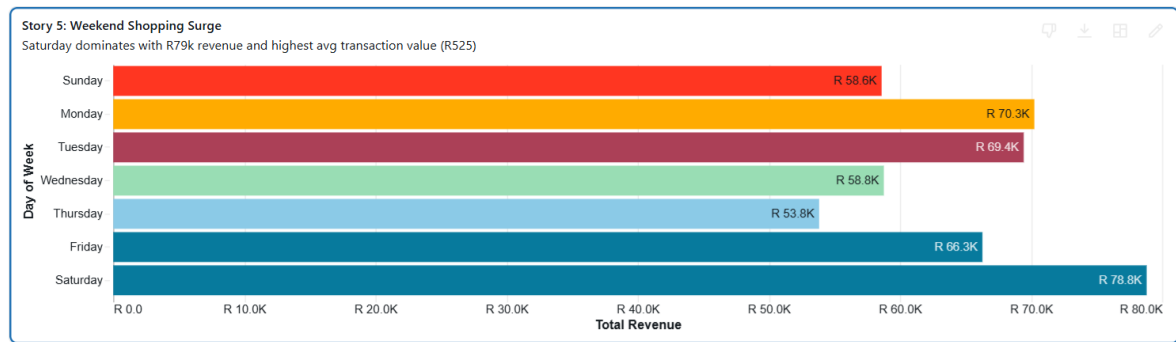


The Insight: **Monthly revenue** swings wildly from R24k (September low) to R53k (May peak) — a 125% range. The grey line shows the R38k average, revealing significant instability. May, October, and December are strong months, while March and September consistently underperform. 4

What This Means: This volatility suggests inconsistent marketing or operational issues. Stabilizing revenue through targeted Q3 promotions and consistent customer acquisition could reduce risk and improve cash flow predictability.

Story 5: Weekend Shopping Surge

Show code

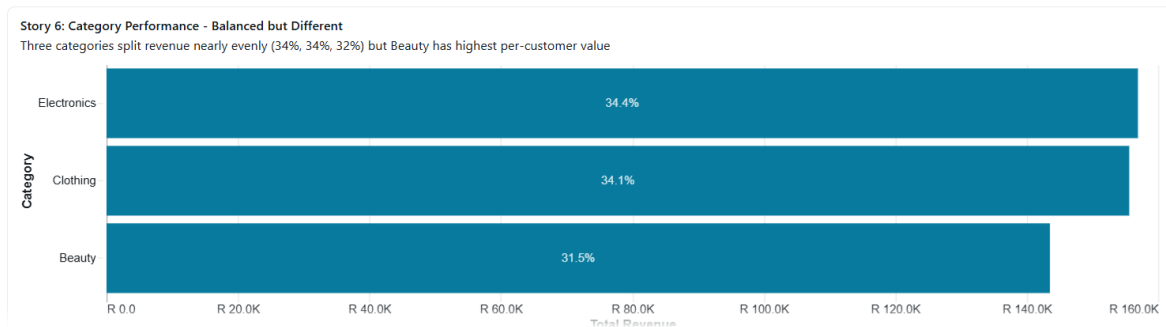


The Insight: **Saturday dominates** with R79k revenue (17.3% of weekly total) and the highest average transaction value at R525. Monday follows at R70k, while Thursday is the weakest day at R54k. 5

What This Means: Weekend shoppers are your most valuable customers. Ensure premium inventory and full staffing on Saturdays. Consider weekend-specific promotions to maximize this natural traffic surge

Story 6: Category Performance - Balanced but Different

Show code

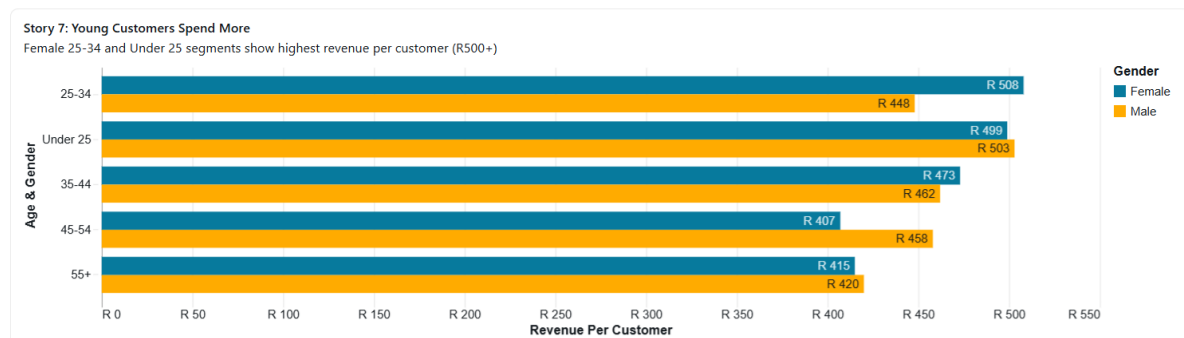


The Insight: **Three categories split revenue nearly evenly** — Electronics (34.4%), Clothing (34.1%), Beauty (31.5%). However, Beauty shows the highest revenue per customer at R467, while Clothing has the most transactions (351). 6

What This Means: No single category dominates, which provides diversification but also means no clear growth leader. Electronics shows 39% monthly growth rate (highest), making it the priority for expansion investment.

Story 7: Young Customers Spend More

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The Insight: **Female 25-34 customers** generate R508 revenue per customer (highest), followed by Male Under 25 at R503. The 55+ and 45-54 age groups show the lowest per-customer values (R407-R420). 7

What This Means: Younger demographics are your most valuable segments despite representing smaller customer counts. Marketing spend should prioritize Female 25-34 and Under 25 segments, particularly for premium products.

The Complete Picture

These seven visual stories reveal a business with strong fundamentals but critical vulnerabilities:

Strengths:

- Premium products and high-value customers drive unequal revenue
- Weekend traffic and larger basket sizes show clear revenue multipliers
- Balanced category performance provides diversification

Critical Risks:

- Zero customer retention (100% one-time buyers)
- Heavy dependence on 20% of customers for 63% of revenue
- Extreme revenue volatility (125% monthly swing)
- 60% of transactions are low-value ($\leq$ R50 items)

Immediate Actions:

- Launch retention program targeting premium customers (20%)
- Implement basket size incentives (3-4 unit bundles)
- Expand premium product lines (currently 54% of revenue)
- Stabilize Q3 performance with targeted September promotions
- Optimize Saturday operations (highest revenue day)

08-WRITE-UP DOC

What are biggest business opportunities identified from the data and which products and categories should management prioritize

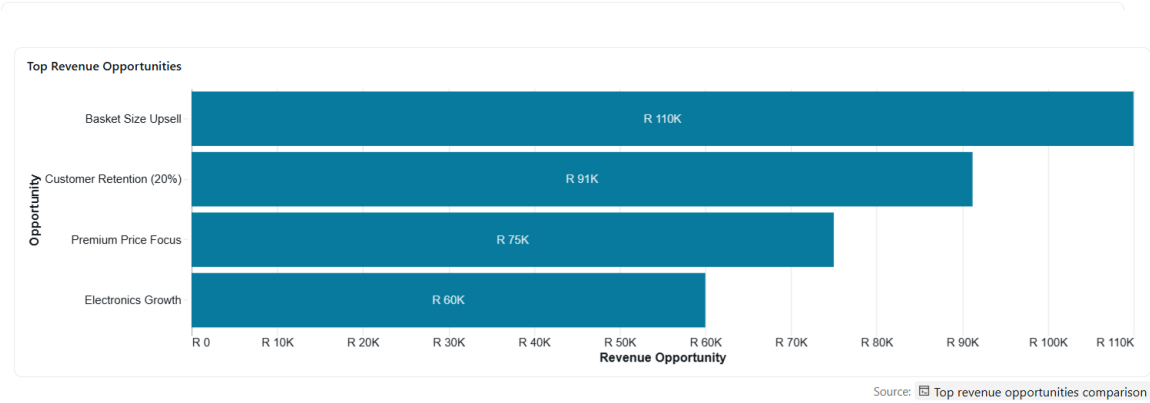
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Biggest Business Opportunities and Product Prioritization

Based on comprehensive analysis of the retail sales data, here are the **top revenue opportunities** ranked by potential impact, along with strategic product and category prioritization recommendations for management.

Top 4 Revenue Opportunities (Ranked by Impact)

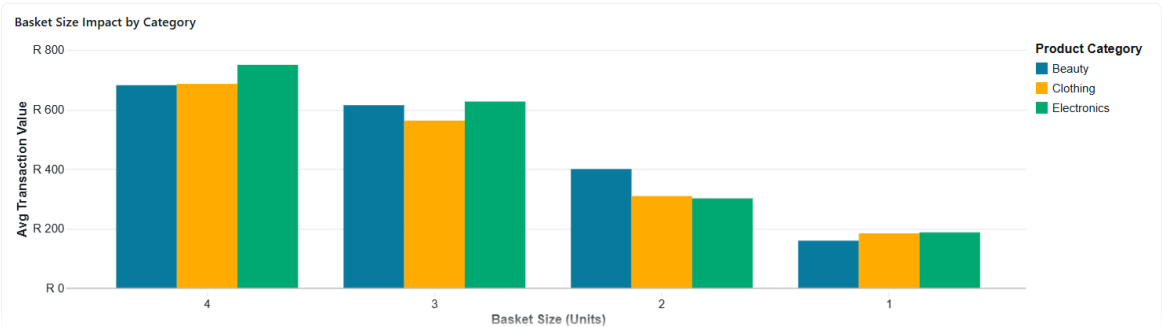
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1. Basket Size Upselling: ~R110k Annual Opportunity

The single largest opportunity lies in increasing basket sizes. Currently, **4-unit purchases generate R749 average transaction value for Electronics**, compared to just R186 for single-unit purchases—a 4x difference.¹

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- Electronics: 4-unit baskets contribute 44.4% of category revenue despite being only 27.2% of transactions
- Clothing: 4-unit baskets contribute 42.7% of revenue from 27.6% of transactions
- Beauty: 3-4 unit baskets combined contribute 70.9% of revenue

Action: Implement bundle offers, “frequently bought together” recommendations, and quantity discounts to shift customers from 1-2 unit purchases to 3-4 units.

2. Customer Retention: R91k Opportunity (20% Retention Rate)

With **100% of customers making only one purchase**, even modest retention improvements yield massive returns. A 20% retention rate would generate R91k in additional annual revenue.²

Action: Launch post-purchase email campaigns, loyalty programs, and personalized offers within 30 days of first purchase.

3. Premium Price Point Focus: ~R75k Opportunity

Premium products (>R300) already generate **54.3% of total revenue** from only 19.9% of transactions. There’s significant headroom to expand premium sales.³

Within categories:

- Beauty: Premium products contribute 58.9% of revenue
- Electronics: Premium products contribute 53.9% of revenue
- Clothing: Premium products contribute 50.5% of revenue⁴

Action: Expand premium product lines, enhance premium product merchandising, and target high-value customer segments (Female 25-34, Under 25) with premium offerings.

4. Electronics Category Growth: ~R60k Opportunity

Electronics shows 38.9% average monthly growth rate—14x higher than any other category. ³ This explosive growth trajectory presents the clearest expansion opportunity.

Product Category Prioritization Matrix

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Product Category Performance Summary						
Category	Total Revenue	Transactions	Revenue/Customer	Avg Transaction	% of Revenue	
Electronics	R 156,905	342	R 459	R 459	34.4	
Clothing	R 155,580	351	R 443	R 443	34.1	
Beauty	R 143,515	307	R 467	R 467	31.5	

• Business recommendations —

Priority 1: Electronics (High Growth + High Revenue)

- Current Performance: R157k revenue (34.4% of total), R459 revenue per customer
- Growth Rate: 38.9% average monthly growth—the clear growth leader
- Strategic Value: Highest transaction value potential (R749 for 4-unit baskets)
- Recommendation: Aggressive expansion. Increase inventory depth, expand product range, and allocate 40% of marketing budget here.

Priority 2: Beauty (High Margin + Premium Affinity)

- Current Performance: R144k revenue (31.5% of total), R467 revenue per customer (highest)
- Growth Rate: 9.1% monthly growth—stable and positive
- Strategic Value: Highest premium product penetration (58.9% of revenue from premium items)
- Recommendation: Premium positioning. Focus on high-margin premium beauty products, target Female 25-34 and Under 25 segments.

Priority 3: Clothing (Stable Base + Volume Leader)

- Current Performance: R156k revenue (34.1% of total), R443 revenue per customer
- Growth Rate: 2.7% monthly growth—slowest but stable
- Strategic Value: Highest transaction volume (351 transactions), strong 4-unit basket performance
- Recommendation: Volume optimization. Focus on basket size expansion and cross-selling with other categories.

High-Value Customer Segment Targeting

Top 3 Segments to Prioritize:

High-Value Customer Segment Targeting

Top 3 Segments to Prioritize:

1. **Female 25-34:** R508 average customer value (highest), 11.4% of revenue from 10.2% of customers [6](#)
 - **Target Products:** Premium Clothing, Electronics
 - **Strategy:** Personalized premium product recommendations
2. **Male Under 25:** R503 average customer value, 8.5% of revenue from 7.7% of customers
 - **Target Products:** Premium Beauty, Electronics
 - **Strategy:** Digital-first marketing, social media engagement
3. **Female Under 25:** R499 average customer value, 7.9% of revenue from 7.2% of customers
 - **Target Products:** Premium Beauty, Clothing
 - **Strategy:** Influencer partnerships, trend-focused campaigns

Seasonal Opportunity Windows

Peak Revenue Months (prioritize inventory and marketing):

- **May:** R53k (highest), driven by Electronics surge (R23k) [7](#)
 - **October:** R47k, strong across all categories
 - **December:** R45k, Electronics-led (R20k)
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Seasonal Opportunity Windows

Peak Revenue Months (prioritize inventory and marketing):

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Weak Months (require intervention):

- **September:** R24k (lowest)—implement targeted promotions
- **March:** R29k—focus on Clothing and Beauty to offset Electronics weakness

Executive Action Plan

Immediate Actions (Next 30 Days):

1. Launch Electronics inventory expansion (target 25% increase)
2. Implement basket size incentives (buy 3+ get 10% off)
3. Deploy post-purchase retention email campaign

Short-Term (90 Days):

1. Expand premium product lines in all categories (target 30% of SKUs)
2. Launch loyalty program with repeat purchase incentives
3. Develop targeted campaigns for Female 25-34 and Under 25 segments

Long-Term (6-12 Months):

1. Build cross-category bundle offerings (Electronics + Beauty, Clothing + Beauty)
2. Establish seasonal inventory planning around May/October/December peaks
3. Develop premium brand partnerships to strengthen high-margin positioning

Expected Impact

Implementing these priorities could generate **R336k** in incremental annual revenue (74% increase);

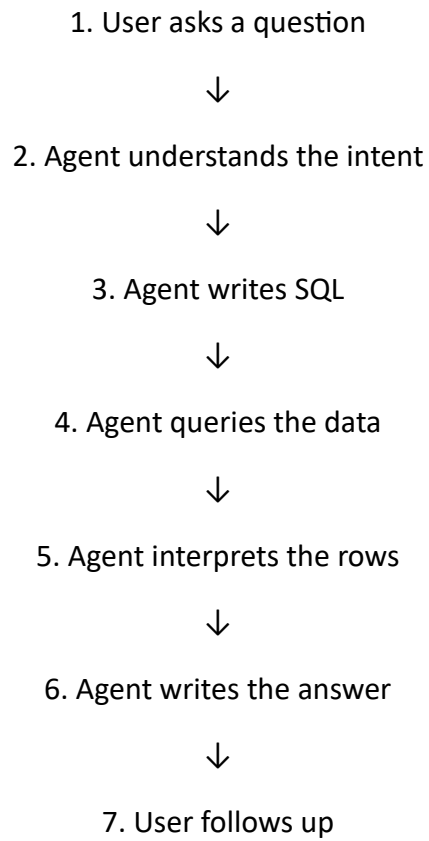
Expected Impact

Implementing these priorities could generate **R336k** in incremental annual revenue (74% increase);

- Basket size optimization: R110k
- Customer retention (20%): R91k
- Premium expansion: R75k
- Electronics growth acceleration: R60k

• **Conclusion** — what worked, what was hard, what you would change.

The Data Agent actually listens to the instructions and gave the outcomes as instructed where The user controls the question and follow-up, while the agent performs the middle analytical steps:



09-PUSH TO GITHUB

- **GitHub repository link**

10-SUBMIT LINK